

CONSTANT OF PROPORTIONALITY – PART 1



LESSON INTRODUCTION

MA.7.AR.4.2 Determine the constant of proportionality within a mathematical or real-world context given a table, graph, or written description of a proportional relationship.

MA.7.AR.4.3 Given a mathematical or real-world context, graph proportional relationships from a table, equation, or a written description.

LEARNING TARGETS

- I can graph a proportional relationship from a table, an equation, or a written description.
- I can determine the constant of proportionality from a graph.
- I can determine the constant of proportionality from a table.

BENCHMARK COHERENCE

BUILDING ON

MA.6.AR.3.2 Given a real-world context, determine a rate for a ratio of quantities with different units. Calculate and interpret the corresponding unit rate.

WORKING TOWARD

MA.8.AR.3.2 Given a table, graph, or written description of a linear relationship, determine the slope.

MA.8.AR.3.4 Given a mathematical or real-world context, graph a two-variable linear equation from a written description, a table, or an equation in slope-intercept form.

ESSENTIAL QUESTIONS

- What is constant of proportionality?
- How can you use a graph or a table to determine a constant of proportionality?

LESSON NARRATIVE

This lesson builds on prior knowledge of using graphs and tables to determine the unit rate of proportional relationships. Students will be introduced to the concept of a constant of proportionality and find the constant of proportionality for multiple real-world relationships. Students extend this knowledge to understand that there are two constants of proportionality that can describe a proportional relationship because proportional relationships can go in either direction. The lesson ends with students reflecting on their understanding of the concepts from the lesson.

COMPONENT SUMMARY

11.2.1 WARM-UP (~5 MIN.)

Sketch the graph of the proportional relationship in context

11.2.3 GUIDED INSTRUCTION (10-15 MIN.)

Find and interpret the constants of proportionality in terms of proportional relationships

11.2.5 YOUR TURN! (5 MIN.)

Independently find and interpret the constant of proportionality given a graph representing a proportional relationship

11.2.2 EXPLORATION (10-15 MIN.)

Explore unit rates in proportional relationships and compare graphs of the same relationship

11.2.4 GUIDED PRACTICE (10 MIN.)

Find and interpret the constant of proportionality given a graph and table representing proportional relationships

11.2.6 WRAP-UP (~5 MIN.)

Reflect on the understanding of constant of proportionality

RESOURCES

GLOSSARY

Key Terms:

- Constant of proportionality

Additional Terms: N/A

MATERIALS

- 11.2.1 Warm-Up video
- Calculator

ADDITIONAL PREPARATION

- 11.2.1 Warm-Up contains videos to accompany the question prompts.
- (Optional) Choose problems from the lesson that correspond to each statement in the 11.2.6 Wrap-Up.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle to understand how a proportional relationship can have two different constants of proportionality.
- Students may struggle to interpret one or both constants of proportionality in context.
- Students may forget to include units in their interpretations of the constant of proportionality.

THINGS TO CONSIDER

- Consider positioning students to discuss question prompts first before writing their answers.
- If time is an issue, then consider assigning 11.2.5 Your Turn! for homework.
- Allowing students to use calculators during the lesson will help students focus on discovery and process of determining both unit rates in proportional relationships. Consider having
- no-calculator questions where students turn their calculator over.

LITERACY AND LANGUAGE ROUTINES

Three Reads

This lesson includes written descriptions of each proportional relationship explored. During partner work, particularly in 11.2.2 Exploration, encourage pairs to engage in a Three Reads routine to ensure they understand the information being presented as well as what they are being asked to do.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD**MA.K12.MTR.7.1 Real-World Contexts**

This lesson provides multiple opportunities for students to explore constant of proportionality in real-world contexts. Support students as they use graphs and tables to connect constant of proportionality and proportional relationships to everyday experiences. Encourage students to think about how the concepts explored in the lesson can be applied to other real-life experiences.

DIFFERENTIATE INSTRUCTION

This lesson introduces the idea of constant of proportionality and the duality of the relationship between variables in proportional relationships. Connecting the language of describing the constant of proportionality and the relationship between variables will be important in encouraging and solidifying student understanding. Emphasize discussion throughout this lesson so students can verbalize these relationships and learn how to describe constant of proportionality accurately.

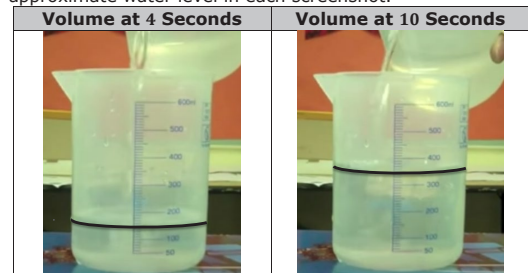
11.2.1 WARM-UP: GRAPHING STORY**ENRICHMENT**

Consider asking scaffolding questions as needed.

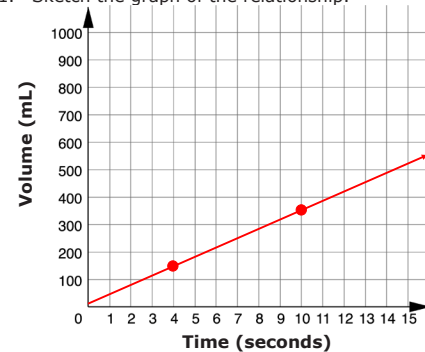
- How can the information from the video be used to sketch a graph?
- How can the information from the screenshots be used to graph the relationship?

**11.2.1 Warm-Up: Graphing Story**

A student is pouring water from a container into an empty beaker. The beaker measures volume in milliliters (mL). The screenshots show the volume of water in the beaker at different points in time. A line is sketched at the approximate water level in each screenshot.



1. Sketch the graph of the relationship.





11.2.2 Exploration: Unit Rates and the Graphs of Proportional Relationships

Work with your partner to complete the following.

- Malcolm went to the gas station to fill up his tank with gasoline. The current price for regular unleaded gasoline at the station is \$2.74 per gallon.

- Determine an approximately equivalent ratio to represent the same relationship per dollar. Round to the nearest thousandth.

$$\frac{\$2.74}{1 \text{ gallon}} \approx \frac{\$1}{0.365 \text{ gallons}}$$

- Both $\frac{\$2.74}{1 \text{ gallon}}$ and $\frac{\$1}{0.365 \text{ gallons}}$ are unit rates. Complete each statement to interpret both unit rates.

- Each gallon of gasoline will cost Malcolm \$ 2.74.
- For every \$1 Malcolm spends, he can add 0.365 gallons to his tank.

- Use either unit rate $\frac{\$2.74}{1 \text{ gallon}}$ or $\frac{\$1}{0.365 \text{ gallons}}$ to complete the table of values to represent this situation. Round to the nearest hundredth when necessary.

Gallons of Gasoline	12	8.30	2.5	5.47 or 5.48*
Cost in Dollars	32.88	22.74	6.85	15

*If students divide 15 by \$2.74, their answer is 5.47 gallons; however, if students multiply 15 by 0.365 then their answer is 5.48, so both are correct.

11.2.2 EXPLORATION: UNIT RATES AND THE GRAPHS OF PROPORTIONAL RELATIONSHIPS

TEACHER MOVES

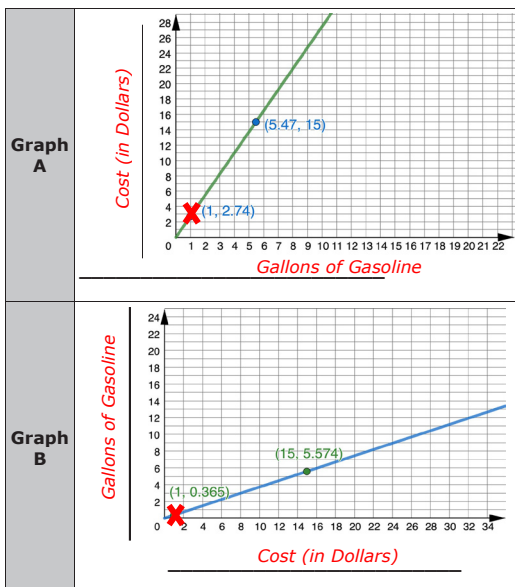
In this section, students consider there are two-unit rates that can describe any proportional relationship, a concept first introduced in Grade 6.

Observe and listen as students work with their partners to determine student thinking to highlight or common misconceptions to address in the next component.

DIFFERENTIATE INSTRUCTION

It may be helpful for students to draw an arrow between rows to indicate which unit rate they are using to fill in the corresponding box. For example, students can draw an arrow pointing from 12 to the empty box and label “ $\times 2.74$ ” to provide an additional visual entry point as they fill in the table.

Two different graphs can be used to represent the proportional relationship between gasoline, in gallons, and cost, in dollars. Both graphs are shown.



- Label the axes of both graphs based on your work in Parts A-C.
Labeled on graph.
- Discuss with your partner which point on each graph represents a unit rate for this relationship. Then, place an X on each graph at this point.
Labeled on graph.

ENRICHMENT

Students may struggle to correctly label the axes for each graph, confusing what is represented on the x- and y-axes. Provide guidance through questioning.

- How can you use the labeled points to determine what the x- and y-values represent on each graph?
- What are the unit rates that describe this proportional relationship?
- How can you use unit rate to determine the labels for each axis?

11.2.3 GUIDED INSTRUCTION: CONSTANT OF PROPORTIONALITY

TEACHER MOVES

Students may not understand that 1.6 kWh each day represents the unit rate. Encourage students to make this connection as they fill in the table and use 1.6 as the unit rate to determine the energy used on days 2 and 3.

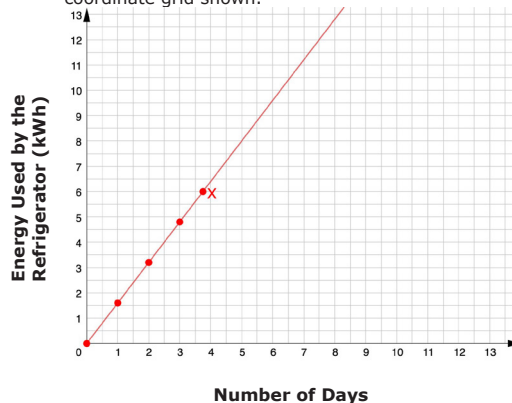


11.2.3 Guided Instruction: Constant of Proportionality

1. Ana's refrigerator uses an average of 1.6 kilowatt-hours (kWh) of energy each day.
 - a. Complete the table of values to represent this relationship.

Number of Days	0	1	2	3
Energy Used (kWh)	0	1.6	3.2	4.8

- b. Plot the ordered pairs from the table on the coordinate grid shown.



- c. Complete the statements to describe the relationship between each point plotted on the graph in Part B.

The y -value ☐ increases ☐ decreases by ☐ 1 unit ☐ 1.6 units every time the x -value ☐ increases ☐ decreases by ☐ 1 unit. ☐ 1.6 units.

- d. Use this information to determine two additional points on the graph of this relationship.
Answers will vary. (5,8) and (10,16)

The graph of the relationship can be completed by using a straightedge to draw a ray with an endpoint at (0,0), going through each point that represents the relationship between the two quantities, with an arrow extending beyond the grid shown.

- e. Complete the graph to represent the relationship by drawing the ray.
See graph.
 - f. Discuss with your partner why the graph of the relationship representing this context has an endpoint at (0,0) and a ray that extends beyond the grid shown.

The graph of a proportional relationship represents an infinite number of points. The point (3.75,6) is included on the graph.

- g. Explain what this point means in terms of the context.
After 3.75 days, the refrigerator has used 6 kilowatt-hours of energy.
- h. Place an X at this point on the graph.
Labeled on graph.



The **constant of proportionality** is the constant value of the ratio of two proportional quantities.

From a graph or a table of values representing a proportional relationship, the constant of proportionality can be determined by expressing any ordered pair as a ratio.



The **constant of proportionality** is the unit rate. It can be determined using the ratio $\frac{a}{b}$ or $\frac{b}{a}$. When two quantities are in a proportional relationship, there are two constants of proportionality, and they are always reciprocals of each other.

11.2.3 GUIDED INSTRUCTION: CONSTANT OF PROPORTIONALITY (CONTINUED)

TEACHER MOVES

While the constant of proportionality leads to slope for linear relationships in Grade 8, it also connects with unit rates that students learned about in Grade 6. It's important that students understand there are two constants of proportionality in any proportional relationship. While it is normally $\frac{y}{x}$ that is used (which later leads to the change in y over the change in x for slope), students need to develop the understanding that proportional relationships can go in either direction. The value changes as well as its units, depending on which direction the relationship is being compared. Sometimes there is a clear independent and dependent variable, while other times there are not.

When modeling how the ratio can be determined between points on the graph of a proportional relationship, it's recommended to show the change in y over the change in x to prime for the introduction to slope in Grade 8, but the axes will be reversed in some cases, like in the 11.2.2 Exploration, to force the understanding that proportional relationships go both ways.

2. Two points on the graph and in the table of values representing the proportional relationship between the number of days and the amount of energy Ana's refrigerator uses are (3, 4.8) and (2, 3.2).

- a. Complete each ratio of the energy used, y, to the number of days, x, for each ordered pair.

$$\frac{y}{x} = \frac{4.8}{3} = \frac{3.2}{2}$$

- b. Determine the value of each ratio using a calculator.
Each ratio is equal to 1.6.

- c. Complete the statement.

The constant of proportionality is the value of each ratio in this relationship, which is 1.6.

- d. Find the reciprocal of the constant of proportionality from Part C.

$$\frac{1}{1.6} = \frac{5}{8} = 0.625$$

- e. Complete the statements.

The other constant of proportionality for the same relationship is _____. In this context, this means

the

- number of days the refrigerator runs
- amount of energy the refrigerator uses

increases by 0.625 units for every 1 unit

increase in the

- number of days.
- amount of energy used.

ENRICHMENT

Students may mix up what the constant of proportionality means in context, especially because we do not typically think of number of days in decimal form. Model student thinking through a Think Aloud.

- How does the reciprocal of the constant of proportionality change the relationship being described? (It reverses the relationship, so it is describing the number of days per unit of energy.)
- How does the number of days change per 1 unit of energy being used based on the constant of proportionality? (It increases by 0.625 day. It takes less than a full day to use 1 unit of energy.)

11.2.4 GUIDED PRACTICE: CONSTANT OF PROPORTIONALITY

ENRICHMENT

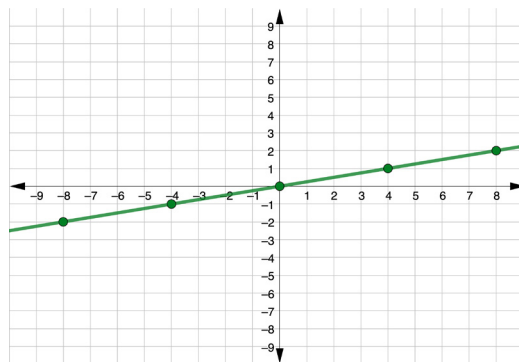
Some students may struggle to find the constant of proportionality using just the graph. Provide further scaffolding through questioning.

- How can you use a point on the graph to help determine constant of proportionality?
- What is the y-value when the x-value is 4? How can that help you determine the constant of proportionality?



11.2.4 Guided Practice: Constant of Proportionality

1. The graph of a proportional relationship is shown.



- a. Determine the constant of proportionality, $\frac{y}{x}$.
The constant of proportionality is $\frac{1}{4}$ or 0.25.
- b. Complete the statement to describe the constant of proportionality.
- The y-value ☐ increases ☐ decreases by 0.25 every time the x-value increases by 1.

TEACHER MOVES

Explaining and understanding proportional relationships can be confusing, particularly when students are first introduced to the concept. Review question 2b out loud with students after they have had an opportunity to try it on their own or with a partner. This will provide an additional auditory entry point, which can help clarify student thinking when making sense of the constant of proportionality in context. As you work through this section, draw connections between the language and the table to illustrate how there are three tablespoons for every one cup of milk.

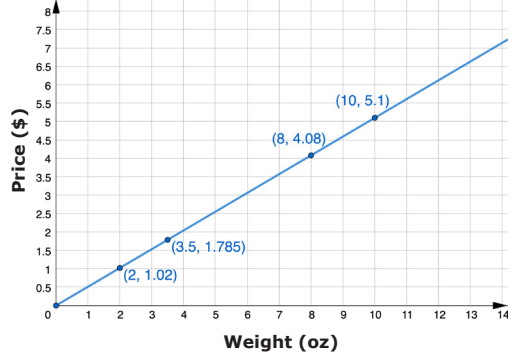
2. The table of values shows how many tablespoons (tbsp) of chocolate milk Dax adds to different quantities of milk to make his preferred flavor of chocolate milk.

Cups of Milk, c	Tablespoons of Chocolate Syrup, T
$\frac{1}{6}$	$\frac{1}{2}$
$\frac{1}{3}$	1
1	3
$1\frac{1}{2}$	4.5
2	6

- a. Determine the constant of proportionality, $\frac{T}{c}$.
The constant of proportionality, $\frac{T}{c}$, is 3.
- b. Complete the statement to interpret this constant of proportionality.
- Dax ☐ increases ☐ decreases the amount of ☐ syrup ☐ milk used by 3 ☐ cups ☐ tablespoons every time the amount of ☐ syrup ☐ milk used increases by ☐ 1 cup. ☐ 1 tbsp.
- c. Determine the constant of proportionality, $\frac{c}{T}$.
The constant of proportionality, $\frac{c}{T}$, is $\frac{1}{3}$.

 11.2.5 Your Turn!

1. At a local frozen yogurt restaurant in Lake Mary, FL, the price of frozen yogurt is determined using its weight, in ounces (oz). The graph shows this relationship.



- a. Determine the constant of proportionality, $\frac{y}{x}$.
The constant of proportionality is 0.51.
- b. Complete the statement to describe the constant of proportionality.

The ☐ price ☐ weight ☐ increases ☐ decreases by 0.51

☐ dollars ☐ ounces every time the ☐ price ☐ weight increases

by ☐ 1 dollar. ☐ 1 ounce.

11.2.5 YOUR TURN!

QUESTIONING

Challenge early finishers to find the price for additional weights of frozen yogurt or to determine the weight based on given prices.

 11.2.6 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each concept after today's lesson.

 I need help.  I got it.  I could help someone.

Understand what the constant of proportionality is in a proportional relationship.	  
Determine the constant of proportionality from a graph.	  
Determine the constant of proportionality from a table.	  

Answers will vary.

11.2.6 WRAP-UP: REFLECTION

DIFFERENTIATE INSTRUCTION

Consider having students label parts of the lesson using the scale provided to illustrate their understanding of the concepts. Choose specific questions ahead of time to have students review as another means of assessment. For example:

- To assess Statement 1, rate yourself on your understanding of 11.2.5 question 1b.
- To assess Statement 2, rate yourself on your understanding of 11.2.5 question 1a.
- To assess Statement 3, rate yourself on your understanding of 11.2.4 question 2a.